

contributions

Context.

- ▶ SVARs are go to models for structural and predictive analyses in macroeconomics
Sims & Zha (2006, AER)
- ▶ Stochastic volatility is the single extension improving forecasts and in-sample fit by the largest margin
Clark & Ravazzolo (2015, JAE), Chan & Eisenstat (2018, JAE)
- ▶ Identification via heteroskedasticity allows flexible estimation and can be verified
Lütkepohl, Shang, Uzeda, Woźniak (2026, JOE)

contributions

Objectives.

To develop a SVAR with new volatility processes that enable verification of identification via heteroskedasticity

Contributions.

- ▶ Sparse heterogeneous Markov-switching for normalised conditional variances
- ▶ New distribution enabling estimation and verification
- ▶ Estimates, normalises, and predicts well
- ▶ Evidence for monetary policy shock identification
- ▶ Coded in the **R** package **bsvars**

heteroskedastic SVARs

heteroskedastic SVARs

Structural VAR.

reduced form: $\mathbf{y}_t = \mathbf{A}\mathbf{x}_t + \boldsymbol{\varepsilon}_t$

structural form: $\mathbf{B}\boldsymbol{\varepsilon}_t = \mathbf{u}_t$

structural shocks: $\mathbf{u}_t \mid \mathbf{x}_t \sim \mathcal{N}(0, \text{diag}(\boldsymbol{\sigma}_t^2))$

covariance: $\boldsymbol{\Sigma}_t = \mathbf{B}^{-1} \text{diag}(\boldsymbol{\sigma}_t^2) \mathbf{B}^{-1'}$

- ▶ heteroskedasticity identifies structural matrix $\mathbf{B}$
- ▶ conditional variances $\boldsymbol{\sigma}_t^2$ need to be normalised

heteroskedastic SVARs

Sparse heterogeneous Markov-switching volatility.

variances: $\sigma_{n,t}^2 = \sigma_{n.S_{n,t}}^2$

Markov process: $s_{n,t} \sim \text{Markov}_{M_n}(\mathbf{P}_n, \boldsymbol{\pi}_n)$

normalisation: $M_m^{-1}(\sigma_{n,1}^2 + \dots + \sigma_{n,M_n}^2) = 1$

prior: $M_m^{-1}(\sigma_{n,1}^2, \dots, \sigma_{n,M_n}^2) \sim \mathcal{D}_{M_n}(\underline{e}_\sigma, \dots, \underline{e}_\sigma)$

- ▶ heterogeneity: equation-specific Markov process
- ▶ sparsity: regimes allowed zero occurrences in-sample
- ▶ normalisation: well-specified structural matrix $\mathbf{B}$

heteroskedastic SVARs

A new inverse gamma-based Dirichlet distribution.

Let $z_m \sim \mathcal{IG2}(s_m, \nu_m)$ for $m \in 1, \dots, M$.

Then $x_m = z_m / (z_1 + \dots + z_M)$ for $m \in 1, \dots, M$ follows:

$$\frac{\Gamma\left(\frac{\sum_{m=1}^M \nu_m}{2}\right)}{\prod_{m=1}^M \Gamma\left(\frac{\nu_m}{2}\right)} \left(\prod_{m=1}^M s_m^{-1}\right) \left[\prod_{m=1}^M \left(\frac{s_m}{x_m}\right)^{\frac{\nu_m+2}{2}}\right] \left(\sum_{m=1}^M \frac{s_m}{x_m}\right)^{-\frac{\sum_{m=1}^M \nu_m}{2}}$$

- ▶ a full conditional posterior for normalised variances
- ▶ easy to sample from maintaining normalisation
- ▶ facilitates identification verification

heteroskedastic SVARs

Identification verification.

homoskedasticity: $\sigma_{n.1}^2 = \dots = \sigma_{n.M_n}^2 = 1$

Bayes factor:
$$SDDR_H = \frac{p(\sigma_{n.1}^2 = \dots = \sigma_{n.M_n}^2 = 1 \mid \mathbf{y})}{p(\sigma_{n.1}^2 = \dots = \sigma_{n.M_n}^2 = 1)}$$

- ▶ partial identification conditions
Lütkepohl, Shang, Uzeda, Woźniak (2026, JOE)
- ▶ a heteroskedastic shock identified with probability 1
- ▶ $SDDR_H$ is easy-to-estimate using the new distribution

heteroskedastic SVARs

Alternative volatility specifications.

- ▶ **HMSH(20)** sparse heterogeneous MSH
- ▶ **HMSH(2)** stationary heterogeneous MSH
- ▶ **MSH(20)** sparse MSH
- ▶ **MSH(2)** stationary MSH
Sims & Zha (2006, AER), Lütkepohl & Woźniak (2020, JEDC)
- ▶ **EXH** exogenous regime changes
Brunnermeier, Palia, Sasry, Sims (2021, AER)
- ▶ **SV** non-centred Stochastic Volatility
Lütkepohl, Shang, Uzeda, Woźniak (2026, JOE)

two MC studies

two MC studies

Estimation under misspecified model.

DGP: $\mathbf{B}\mathbf{y}_t = \mathbf{u}_t, \quad \mathbf{u}_t \sim \mathcal{N}_2(\mathbf{0}_2, \text{diag}(\sigma_t^2))$

Generate 100 synthetic time series for each sample size $T \in \{260, 780\}$ and volatility process

- ▶ SV
- ▶ GARCH
- ▶ MSH(2)
- ▶ HMSH(2)

Fit all models listed before.

two MC studies

Estimation efficiency of structural matrix $\mathbf{B}$.

- ▶ Sparse MSH models estimate $\mathbf{B}$ most precisely
- ▶ HMSH models also perform well
- ▶ new models only rivalled by SV fitted to SV DGP in large samples

Results based on RMSE.

two MC studies

Normalisation of conditional variances σ_t^2 .

- ▶ Sparse MSH models normalise variances adequately
- ▶ HMSH models also perform well
- ▶ new models only rivalled by SV and EXH in large samples

Results based on RMSE.

two MC studies

Assessing the identification verification procedure.

- ▶ All models except for stationary MSH detect homoskedasticity well
- ▶ Sparse MSH models are best in detecting heteroskedasticity
- ▶ Our new models verify identification as well or better than SV model by Lütkepohl, Shang, Uzeda, Woźniak (2026, JOE)

Based on rejection rates in I - and q -value approaches.

empirical evidence

forecasting performance

Updated data by Brunnermeier, Palia, Sasry, Sims (2021, AER)

$$\mathbf{y}_t = \begin{bmatrix} ip_t & p_t & R_t & hhc_t & bc_t & GZ_t & ES_t & TS_t & m_t & pcm_t \end{bmatrix}$$

Assessing forecasting performance.

- ▶ All new models better than EXH
- ▶ All new models outperform SV in point forecasting
- ▶ Sparsity more important than heterogeneity in density forecasting

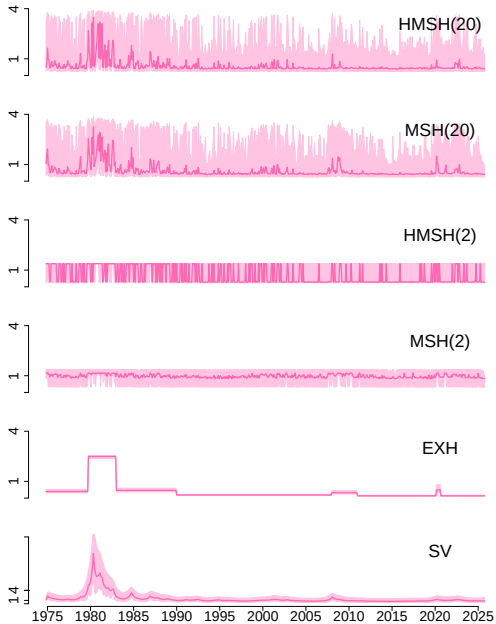
Based on PLS, RMSFE, and MAFE.

identification verification

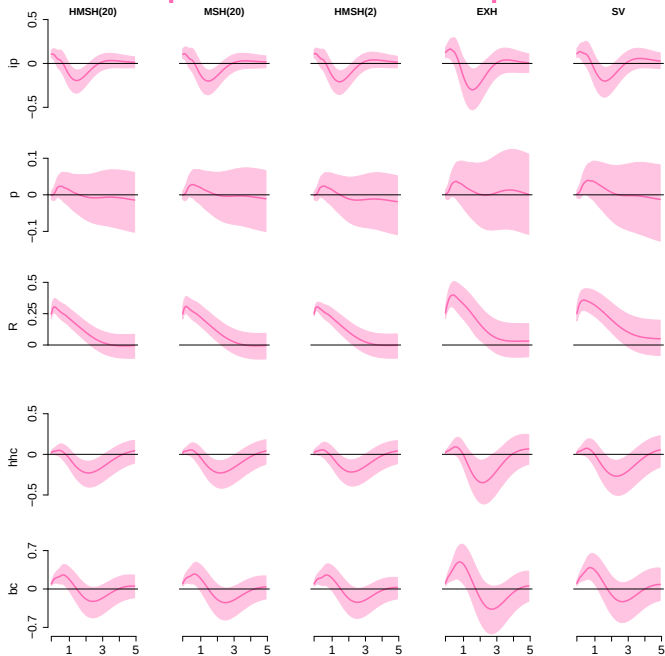
Verify the identification of the US monetary policy shock (mps)

Model	$\ln SDDR_H$
HMSH(20)	-240.283
MSH(20)	-118.314
HMSH(2)	-235.684
MSH(2)	-1.245
EXH	-54.291
SV	-542.583

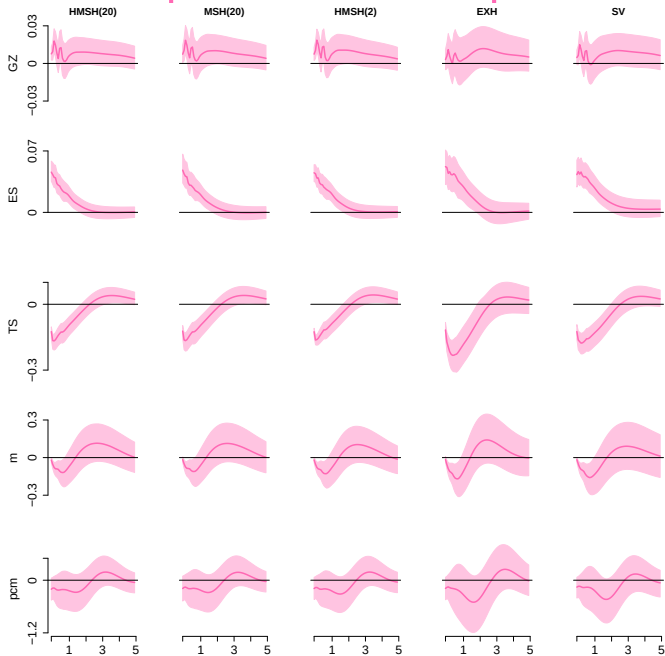
mps conditional variances



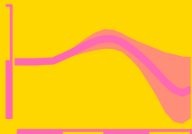
impulse responses to mps



impulse responses to mps



bsvars.org



github.com/donotdespair



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